why are there 4 Bell states?

$$\begin{cases} 1Q \to 2^1\text{-dim Vec.} \\ nQ \to 2^n\text{-dim vector space.} \end{cases}$$

$$|\psi_{1Q}\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \qquad |\psi_{2Q}\rangle = \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix}$$

Spooky Action @ a distance.

There

$$\begin{tikzcd} \text{Alice} & & \text{Bob.} \\ & |\phi\rangle \arrow[ul] \arrow[ur] & \end{tikzcd}$$

what is the essence of quantum entanglement.

↳ Correlations in arbitrary basis.

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{2\sqrt{2}}\Big[(|+\rangle + |-\rangle) \otimes (|+\rangle + |-\rangle) + (|+\rangle - |-\rangle) \otimes (|+\rangle - |-\rangle)\Big]$$

$$= \frac{1}{2\sqrt{2}}\Big[2|++\rangle + 2|--\rangle + \cancel{|+-\rangle} - \cancel{|+-\rangle}\Big]$$

$$= \frac{1}{\sqrt{2}}\Big[|++\rangle + |--\rangle\Big]$$

	A	B
Z-measurement: perfectly correlated.	+1	+1
	−1	−1
X-measurement: perfectly correlated	+1	+1
	−1	−1

$$|00\rangle = \frac{1}{2}(|+\rangle + |-\rangle) \otimes (|+\rangle + |-\rangle) = \frac{1}{2}\left[|++\rangle + |--\rangle + |+-\rangle + |-+\rangle\right]$$

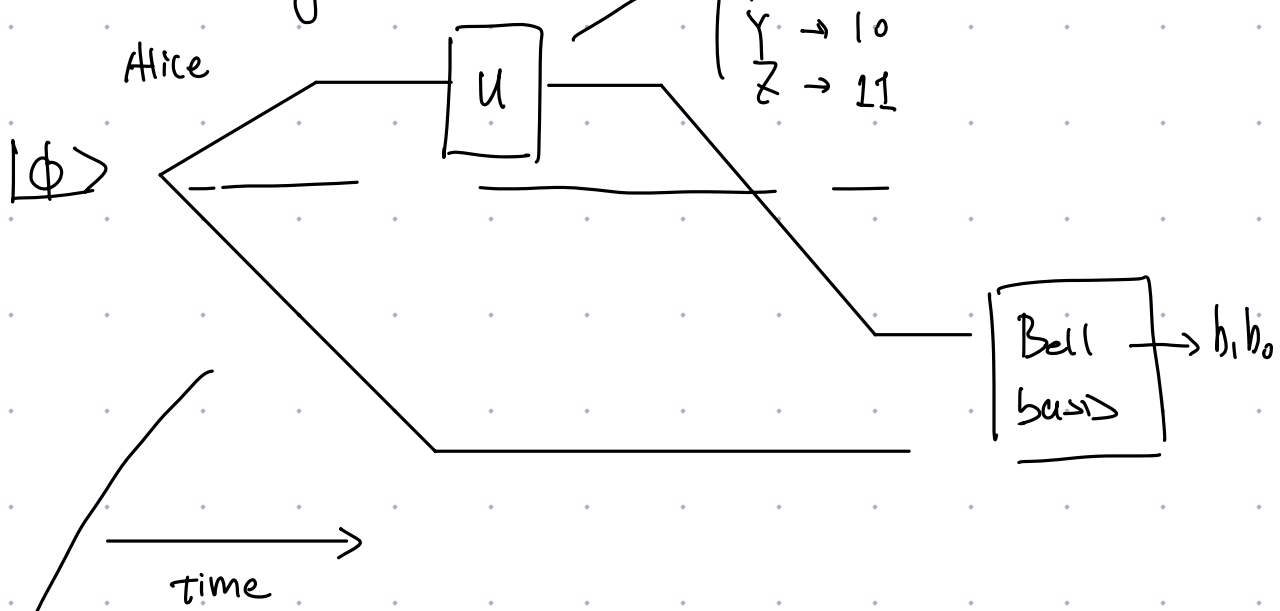
perfectly correlated in the Z-basis

$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
25%	25%	25%	25%
$(+1,+1)$	$(-1,-1)$	$(+1,-1)$	$(-1,+1)$

no correlation in the X basis.

Q-dense Coding.

$$\begin{cases} I \to 00 \\ X \to 01 \\ Y \to 10 \\ Z \to 11 \end{cases}$$



Protocol that uses entanglement that allows you to use 1 qubit to convey 2 bits of information.

Holevo's Bound: n qubits $\rightsquigarrow$ at most n bits of information

entanglement = resource. $\longrightarrow$ bits, qubits, entanglement.

↳ thermodynamics.

they are all resources & information $\Rightarrow$ you cannot forget entanglement as a source of info $\rightarrow$ leads to paradox

Problem?

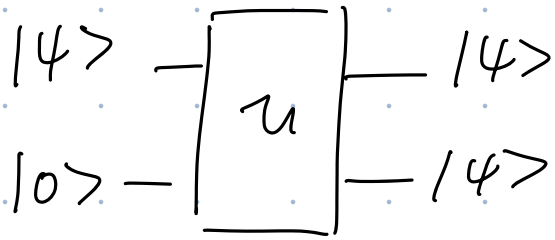
→ Noise :(Entanglement is fragile), can destroy entanglement.

No Cloning Theorem (Wootters, Żurek)

↳ Given an arbitrary state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

there does not exist a unitary U, s.t. $U(|\psi\rangle \otimes |0\rangle)$

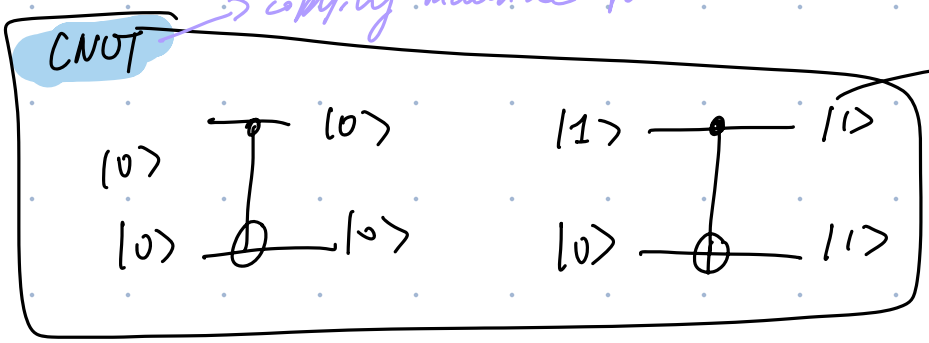
perfectly distinguishable



but this is not arbitrary

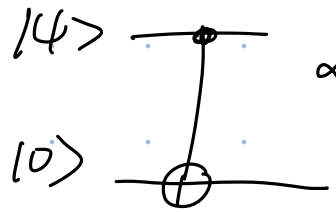
→ copying machine for states |0> or |1>

CNOT



→ this is like a copy-machine.

what about



$\alpha|00\rangle + \beta|11\rangle$

$\neq$

$$\text{CNOT}\left(\alpha|0\rangle + \beta|1\rangle\right) =$$

$$|\psi\rangle \otimes |\psi\rangle = \alpha^2|00\rangle + \alpha\beta|01\rangle + \beta\alpha|10\rangle + \beta^2|11\rangle$$

This is an example, not a proof.

either $\alpha, \beta = 0$

$\Downarrow$

$$\underbrace{|\psi\rangle = 1 \text{ or } |\psi\rangle = 0}$$

Suppose U exists,

$|\psi\rangle, |\phi\rangle \rightarrow$ <u>arbitrary states</u>

← global phase

$$\begin{cases} U(|\psi\rangle \otimes |\phi\rangle) = e^{i\varphi(\psi)} |\psi\rangle \otimes |\psi\rangle \\ U(|\phi\rangle \otimes |0\rangle) = e^{i\varphi(\phi)} |\phi\rangle \otimes |\phi\rangle \end{cases}$$

$\Rightarrow$ 2 states, you can take the inner product:

$$e^{i(\varphi(\psi) - \varphi(\phi))} (\langle\phi| \otimes \langle\phi|)(|\psi\rangle \otimes |\psi\rangle)$$

$$= e^{i(\varphi(\psi) - \varphi(\phi))} (\langle\phi|\psi\rangle) \otimes (\langle\phi|\psi\rangle)$$

$$= e^{i\#} \langle\phi|\psi\rangle \cdot \langle\phi|\psi\rangle = e^{i\#} \langle\phi|\psi\rangle^2$$

$\|$ should equal

$$(U(|\phi\rangle \otimes |0\rangle))^\dagger (U(|\psi\rangle \otimes |0\rangle))$$

$$= (\langle\phi| \otimes \langle 0|) \underbrace{\cancel{U^\dagger U}}_{I} (|\psi\rangle \otimes |0\rangle)$$

$$= \langle\phi|\psi\rangle \cdot \langle 0|0\rangle$$

$$= \langle\phi|\psi\rangle$$

Hence $\langle\phi|\psi\rangle = \langle\phi|\psi\rangle^2$

this can only be true either $\langle\phi|\psi\rangle = 0$,

or $\langle\phi|\psi\rangle = 1$ $(|\phi\rangle = e^{i\varphi}|\psi\rangle)$

$\Rightarrow$ $\boxed{|\phi\rangle \text{ cannot be arbitrary}}$ $\square$.

Alternative way:

$$U\left[\left(\alpha|0\rangle + \beta|1\rangle\right)\otimes|0\rangle\right]$$

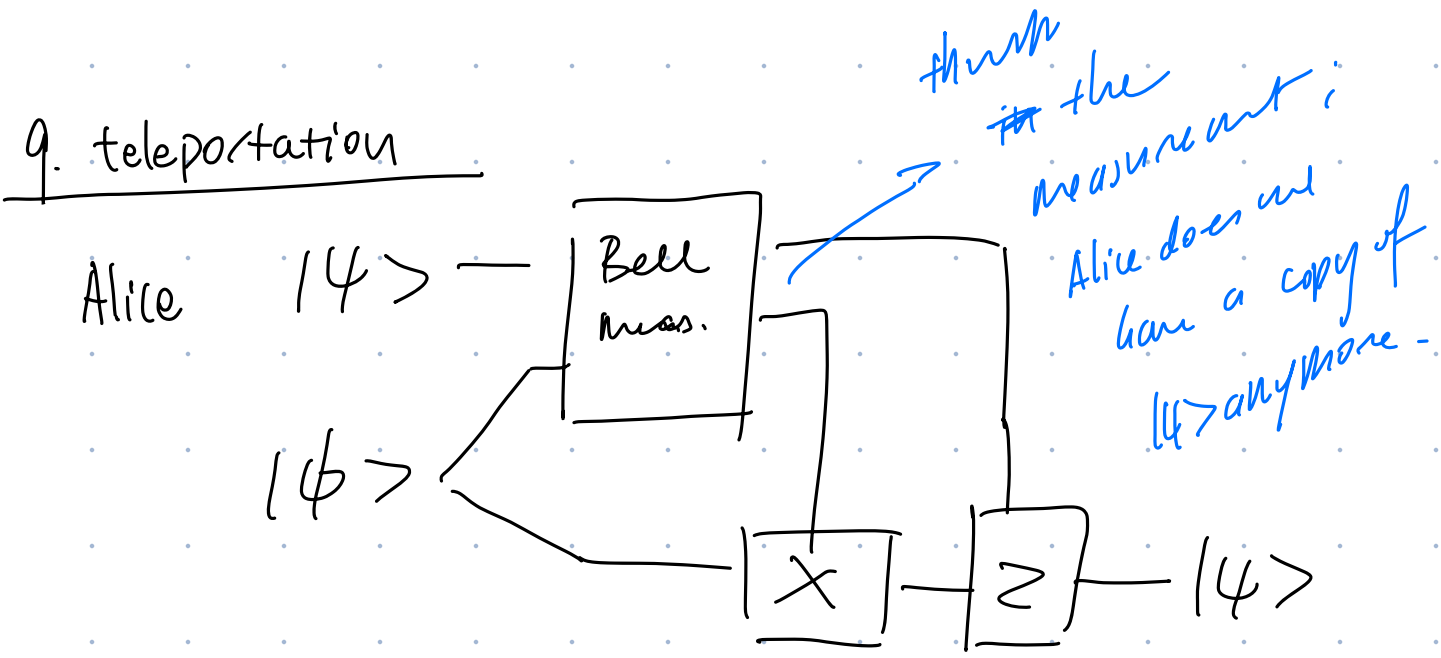
$$= U\left(\alpha|00\rangle + \beta|01\rangle\right)$$

output

$$|\psi\rangle\otimes|\psi\rangle = \underbrace{\alpha^2|00\rangle + \alpha\beta|01\rangle + \beta\alpha|10\rangle + \beta^2|11\rangle}$$

U - if α, β independent, then it is a linear map.

but here we have non-linearity

$\Rightarrow$ Hence U cannot be α, β independent.

9. teleportation